Project Documentation

C64 Keyboard 3D printables

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C64 Mechanical Keyboard & Kernal Switcher

3D Printables

Module Description

1. Introduction

The C64 Keyboard PCB requires either a **mounting frame** for installation in a C64 or C64C case, or a **USB keyboard case**. All parts were designed to fit on a 3D printer with a **25 cm-wide print bed**.

The mounting frame has been tested with various cases, including the **C64 Breadbin**, different **C64C** cases, and the **C64C clip case**.

There are two different versions of the C64 USB keyboard:

- **Classic:** For use with original keycaps and a classic 9U space bar.
- **Modern:** For use with modern keycaps, including DIY and Blingboard64/C64U keycaps.

The difference between the two versions is the **plates**.

2. The *Mounting* Frame

The mounting frame is attached to the C64 Keyboard PCB using **16 C 2.2 × 6.5 H DIN 7981 self-tapping sheet-metal screws**.

Figure 1: Mounting frame (Rev. 1)

For **PCB Rev. 0**, a **Rev. 0 mounting frame** is required. For **PCB Rev. 1 and Rev. 2**, a **Rev. 1 mounting frame** is required. The difference is that the Rev. 1 mounting frame supports a **9U stabilizer**, which requires one of the mounting screws to be moved a few millimeters to the right.

A **stiffener** is an optional part used to increase the rigidity of the PCB. It is attached to the keyboard using **9 C 2.2 × 6.5 H DIN 7981 self-tapping sheet-metal screws** (Figure 4). In my experience, the stiffener is not required. It is identical for all PCB revisions.

There are two additional **C 2.9 × 9.5 H DIN 7981** screws that hold the two sides of the frame together. They are installed approximately in the middle of the **north and south sides**, as shown in Figure 2.

Figure 2: the two halves of the frame, joined by two horizontal screws

Figure 3: The keyboard PCB Rev. 0 mounted on a Rev. 0 frame

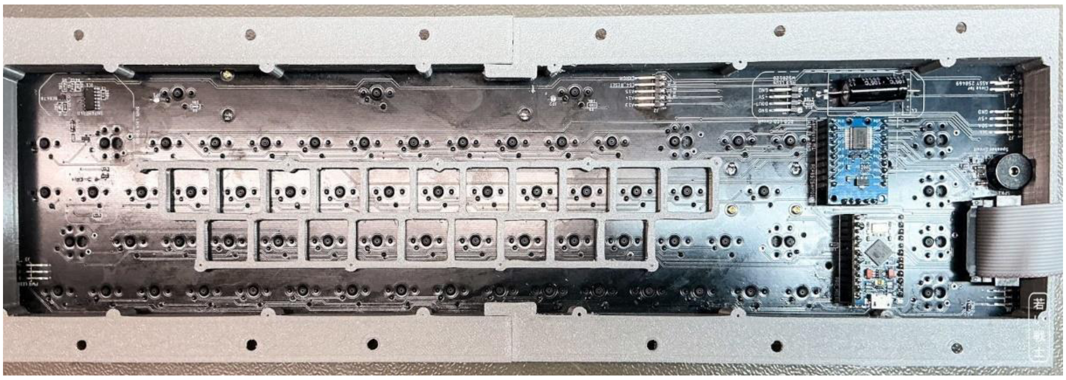

Figure 4: Keyboard with stiffener

Figure 5: Keyboard mounted in a C64 breadbin case (clear, RetroFuzion)

In a C64C clip case, the keycaps protrude slightly from the case. This is because the PCB needs to rest against the case. The height of the mounting frame is determined by the height of the clips.

With a standard C64C case, the height of the keycaps is perfectly acceptable.

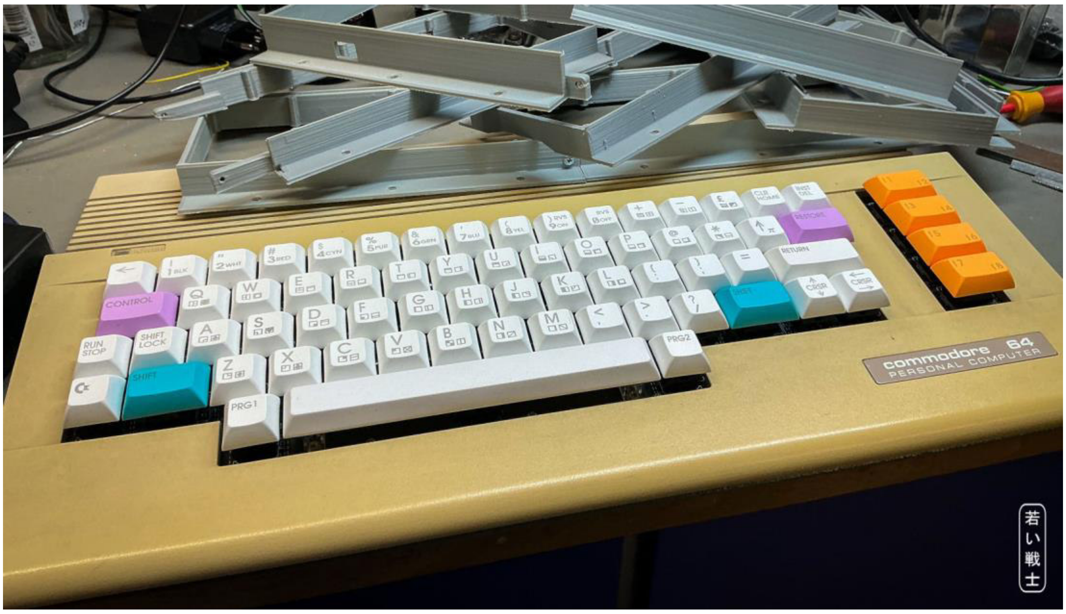

Figure 6: Keyboard in a C64C clip case

Figure 7: The clip case clip, the C 2.9x6.5 H screw

Figure 8: C64 keyboard in a normal C64C case

3. The C64 USB Keyboard Case

The USB keyboard case consists of **7 different parts**:

- **The plate** (standard or classic) – two halves
- **The case** – two halves
- **The bottom** – 3 parts

The plate is elevated **5 mm** above the PCB and has cutouts for the keyboard switches. To fit on an average 3D printer, it is printed in two parts.

There are two versions:

- **Standard**: all 1.5U switches populated in **SW**A** positions, **7U stabilizer**

- **Classic**: all 1.5U switches populated in **SW**** positions, **9U stabilizer**

The case is divided into two halves for the same reason: to fit on a standard 3D printer.

The bottom is divided into three parts. The middle part spans over both case halves to add some stability.

The USB keyboard case requires **42 C 2.2 × 6.5 H DIN 7981 self-tapping sheet-metal screws**:

- 16 for the PCB
- 14 for the plate
- 12 for the bottom

Figure 9: USB-Keyboard case parts: plate (top), case (mid), bottom (bottom)

It is important that the plates are properly **de-burred** before installation. They are already slightly wider than the standard dimensions because cutouts tend to be narrower than designed when 3D printed.

Failing to de-burr the plates can result in excessive force being required to install the switches, switches not seating properly—especially when using **hot-swap sockets**—and damage to the hot-swap sockets. I have experienced the latter myself.

Figure 10: USB keyboard case with DIY keycaps (UV-decals)

Figure 11: USB keyboard case with original keycaps

The plates would also fit with the mounting frame, if desired. They are made for PCB Rev. 1 and 2. They are using the same mounting holes of the PCB like the stiffener (plus holes added in PCB Rev. 1).